Trainings

General Information

All maintenance procedures listed for previous intervals **must** also be performed.

Perform maintenance at whichever interval occurs first. At each scheduled maintenance interval, perform all previous maintenance checks that are due for scheduled maintenance.

For convenience, listed below are the section numbers that contain specific instructions for performing the maintenance.

The K38, K50, QSK38 and QSK50 engines have the following lubricating oil filtration options:

Marine and Industrial

- Spin-on Oil Filter without Centrifuge and without Centinel™
- Spin-on Oil Filter with Centrifuge and without Centinel™
- Spin-on Oil Filter without Centrifuge and with Centinel™
- Spin-on Oil Filter with Centrifuge and with Centinel™
- Eliminator™ without Centinel™
- Eliminator™ with Centinel™

Power Generation

- Spin-on Oil Filter without Centrifuge and without Centinel™
- Spin-on Oil Filter with Centrifuge and without Centinel™
- Spin-on Oil Filter without Centrifuge and with Centinel™
- Spin-on Oil Filter with Centrifuge and with Centinel™
- Eliminator™ without Centinel™
- Eliminator™ with Centinel™

Locomotive

- Eliminator™ with Centinel™
- Eliminator™ without Centinel™

The following applicable maintenance schedule for the engine application and the specific oil filtration option used.

Marine and Industrial

Spin-on Oil Filter without Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Lubricating Oil and Filters - Change[7]
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check

- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Air Shutoff Valve - Inspect for Reuse
- Turbocharger – Check (All Applications except Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude
- Turbocharger – Replace (Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude

Spin-on Oil Filter with Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check

- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Lubricating Oil and Filters - Change[7]
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Fleetguard® Centrifuge - Clean[7]
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Air Shutoff Valve - Inspect for Reuse
- Turbocharger – Check (All Applications except Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude
- Turbocharger – Replace (Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude

Spin-on Oil Filter without Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]

- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubricating Oil Filters - Change[7]

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Air Shutoff Valve - Inspect for Reuse
- Turbocharger – Check (All Applications except Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude

- Turbocharger – Replace (Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude

Spin-on Oil Filter with Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Lubricating Oil Analysis - Check[7]
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- Charge Air Cooler - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubricating Oil Filters - Change[7]

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Fleetguard® Centrifuge Filter - Clean[7]
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Air Shutoff Valve - Inspect for Reuse
- Turbocharger – Check (All Applications except Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude
- Turbocharger – Replace (Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude

Eliminator™ without Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check

- Centinel™ Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Eliminator™ Filter Rotation - Check
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 250 Hours or 1 Year

- Lubricating Oil and Filters - Change[7]

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean and Check[7]
- Eliminator™ Filter Pressure - Check

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check

- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Air Shutoff Valve - Inspect for Reuse
- Turbocharger – Check (All Applications except Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude
- Turbocharger – Replace (Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude

Eliminator™ with Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Eliminator™ Filter Rotation - Check
- Centinel™ Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check

- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean[7]
- Eliminator™ Filter Pressure - Check

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]

- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Air Shutoff Valve - Inspect for Reuse
- Turbocharger – Check (All Applications except Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude
- Turbocharger – Replace (Non-Regulated (Tier 0) Single Stage Turbocharged Haul Truck Applications operating above 2740 meters [9000 feet] altitude

The below list is in reference to the upper indexes used in above Marine and Industrial scheduled maintenance.

1. Mechanically actuated injector engines **only**.
2. Electronically actuated injector engines **only**. Alternative fuel filter change information is available in the fuel filter change interval section of this procedure.
3. After an initial adjustment at 1500 hours, it is recommended that the valves and injectors **not** be adjusted again prior to injector calibration at the 6000 hours or 2 years interval.
4. Calibration **must** be performed by the nearest Cummins® Authorized Repair Location.
5. Recommended at engine half life to rebuild for Cummins® Dual Fuel Kit equipped engines. Half life varies by application. Contact the nearest Cummins® Authorized Repair Location if the interval can **not** be determined.
6. CENSE™ CM530 modules **only.** Reset is **not** necessary on CM2330 modules.
7. See Service Bulletin, Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html), and Service Bulletin, Extended Service Interval Program for Generator Set Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html).

Power Generation

Spin-on Oil Filter without Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2.7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 250 Hours or 1 Year

- Lubricating Oil and Filters - Change[7,8]

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check

- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Crankcase Breather Element - Change
- Air Shutoff Valve - Inspect for Reuse

Spin-on Oil Filter with Centrifuge and without Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 250 Hours or 6 months

- Lubricating Oil and Filters - Change[7,8]

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Fleetguard® Centrifuge - Clean[7,8]
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Crankcase Breather Element - Change
- Air Shutoff Valve - Inspect for Reuse

Spin-on Oil Filter without Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 Months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7]- Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubrication Oil and Filters - Change[7,8]

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Crankcase Breather Element - Change
- Air Shutoff Valve - Inspect for Reuse

Spin-on Oil Filter with Centrifuge and with Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check

- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Lubricating Oil Analysis - Check
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- Charge Air Cooler - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Lubricating Oil Filters - Change[7,8]

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Fleetguard® Centrifuge Filter - Clean[7,8]
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check

- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Crankcase Breather Element - Change
- Air Shutoff Valve - Inspect for Reuse

Eliminator™ without Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Eliminator™ Filter Rotation - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain

- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 250 Hours or 1 Year

- Lubricating Oil and Filters - Change[7,8]

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean[7,8]
- Eliminator™ Filter - Pressure Check

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]

- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]
- Vibration Damper, Viscous - Check
- Crankcase Breather Element - Change
- Air Shutoff Valve - Inspect for Reuse

Eliminator™ with Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Coolant Level - Check
- Sea Water Strainer - Clean
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs - Drain
- Marine Gear - Check
- Drive Belts - Check
- Fuel Filter, Remote-Mounted[2] - Drain
- Eliminator™ Filter Rotation - Check
- Fuel Filter, Remote-Mounted[2] - Check

Maintenance Procedures at 250 Hours or 6 months

- Fuel Filter (Spin-On Type)[1,7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check

- Charging System Alternator - Clean
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Zinc Anode - Check
- CENSE™ Datalogger - Reset[6]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter - Check

Maintenance Procedures at 1000 Hours or 1 Year

- Eliminator™ Filter Centrifuge - Clean[7,8]
- Eliminator™ Filter - Pressure Check

Maintenance Procedures at 1500 Hours or 1 Year

- Engine Steam Cleaning - Clean
- Overhead Set Outer Base Circle (OBC) - Adjust[1,3]
- Overhead Set (Travel Method) - Adjust[1,3]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Fan Drive Idler Arm Assembly - Check
- Radiator Hoses - Check
- Cooling Fan Belt Tensioner - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 6000 Hours or 2 Years

- Fuel Pump - Calibrate[1,4]
- Cooling System - Flush[7]
- Fan Hub, Belt Driven - Check
- Sea Water Pump - Check
- Water Pump - Check
- Air Compressor Unloader and Valve Assembly - Check

Maintenance Procedures at 6000 Hours

- Injector - Calibrate[1,4]
- Turbocharger - Inspect for Reuse

Maintenance Procedures at 10,000 Hours

- Injectors - Replace[2,5]

- Vibration Damper, Viscous - Check
- Crankcase Breather Element - Change
- Air Shutoff Valve - Inspect for Reuse

The below list is in reference to the upper indexes used in above Power Generation scheduled maintenance.

1. Mechanically actuated injector engines **only**.
2. Electronically actuated injector engines **only**. Alternative fuel filter drain information is available in the fuel filter section of this procedure.
3. After an initial adjustment at 1500 hours, it is recommended that the valves and injectors **not** be adjusted again prior to injector calibration at the 6000 hours or 2 years interval.
4. Calibration **must** be performed by the nearest Cummins® Authorized Repair Location.
5. Recommended at engine half life to rebuild for Cummins® Dual Fuel Kit equipped engines. Half life varies by application. Contact the nearest Cummins® Authorized Repair Location if the interval can **not** be determined.
6. CENSE™ CM530 modules **only:** Reset is **not** necessary on CM2330 modules.
7. See Service Bulletin, Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html) and Service Bulletin, Extended Service Interval Program for Generator Set Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html).
8. Lubricating oil change interval can be can be extended to 2 years if oil does **not** exceed designated hours and oil is sampled quarterly with oil analysis parameters remaining within the limits specified in Table 2 of the Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060. (/qs3/pubsys2/xml/en/bulletin/4022060.html)

Locomotive

Eliminator™ with Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Centinel™ Oil Level - Check
- Coolant Level – Check
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs – Drain
- Eliminator™ Filter Rotation - Check
- Drive Belts – Check

Maintenance Procedures at 250 Hours or 14 Days

- Lubricating Oil Analysis - Check[7]

Maintenance Procedures at 1500 Hours or 92 Days

- Fuel Filter (Stage 0)[7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Fan, Cooling - Check
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter – Check
- Exhaust Manifold (Dry) – Check
- Lubricating Oil Filters - Change

Maintenance Procedures at 6000 Hours or 368 Days

- Engine Steam Cleaning – Clean
- Crankshaft - Check
- Overhead Set Outer Base Circle (OBC) - Adjust[3]
- Overhead Set (Travel Method) - Adjust[3]
- Lubricating Oil - Change[7]
- Engine Oil Heater - Check
- Coolant Heater - Check
- Radiator Hoses - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 10000 Hours or 1104 Days

- Vibration Damper, Viscous – Check
- Cooling System - Flush[7]
- Water Pump – Check
- Thermostats – Replace
- Injectors – Replace
- Turbocharger – Replace
- Aftercooler – Check
- Crankcase Blowby, Measure – Check
- Exhaust Manifold, Dry – Check
- Exhaust Connection Pipe – Check

- Bellows - Check
- Wiring Harness – Check
- Fuel Supply Lines – Check
- Injector Supply Lines (High Pressure) – Check
- Air Compressor Unloader and Valve Assembly - Check

Locomotive

Eliminator™ without Centinel™

Maintenance Procedures at Daily Interval

- Crankcase Breather Tube - Check
- Fuel-Water Separator - Drain
- Lubricating Oil Level - Check
- Coolant Level – Check
- Air Cleaner Precleaner - Clean
- Air Intake Piping - Check
- Air Cleaner Restriction - Check
- Air Tanks and Reservoirs – Drain
- Eliminator™ Filter Rotation - Check
- Drive Belts - Check

Maintenance Procedures at 1500 Hours or 92 Days

- Fuel Filter (Stage 0)[7] - Change
- Fuel Filter (Stage 1)[2,7] - Change
- Fuel Filter (Stage 2)[2,7] - Change
- Coolant Filter - Change[7]
- Supplemental Coolant Additive (SCA) and Antifreeze Concentration - Check[7]
- Lubricating Oil and Filters - Change[7]
- Engine Support Bracket, Front - Check
- Engine Mounts - Check
- Water Pump Weep Hole Filter – Check
- Exhaust Manifold (Dry) – Check

Maintenance Procedures at 6000 Hours or 368 Days

- Engine Steam Cleaning – Clean
- Crankshaft - Check
- Overhead Set (OBC) - Adjust[3]
- Overhead Set (Travel Method) - Adjust[3]

- Engine Oil Heater - Check
- Coolant Heater - Check
- Radiator Hoses - Check
- Air Compressor Discharge Lines - Check
- Batteries - Check
- Battery Cables and Connections - Check

Maintenance Procedures at 10000 Hours or 1104 Days

- Vibration Damper, Viscous – Check
- Cooling System - Flush[7]
- Water Pump – Check
- Thermostats – Replace
- Injectors – Replace
- Turbocharger – Replace
- Aftercooler – Check
- Crankcase Blowby, Measure – Check
- Exhaust Manifold, Dry – check
- Exhaust Connection Pipe – Check
- Bellows - Check
- Wiring Harness – Check
- Fuel Supply Lines – check
- Injector Supply Lines (High Pressure) – Check
- Air Compressor Unloader and Valve Assembly – Check

The below list is in reference to the upper indexes used in above Locomotive scheduled maintenance.

1. Mechanically actuated injector engines **only**.
2. Electronically actuated injector engines **only**. Alternative fuel filter change information is available in the fuel filter change interval section of this procedure.
3. After an initial adjustment at 1500 hours, it is recommended that the valves and injectors **not** be adjusted again prior to injector calibration at the 6000 hours or 2 years interval.
4. Calibration **must** be performed by the nearest Cummins® Authorized Repair Location.
5. Recommended at engine half life to rebuild for Cummins® Dual Fuel Kit equipped engines. Half life varies by application. Contact the nearest Cummins® Authorized Repair Location if the interval can **not** be determined.
6. CENSE™ CM530 modules **only:** Reset is **not** necessary on CM2330 modules.
7. See Service Bulletin, Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html) and Service Bulletin, (/qs3/pubsys2/xml/en/bulletin/5411272.html)Extended Service Interval Program for Generator Set Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html).

Oil and Oil Filter Change Interval

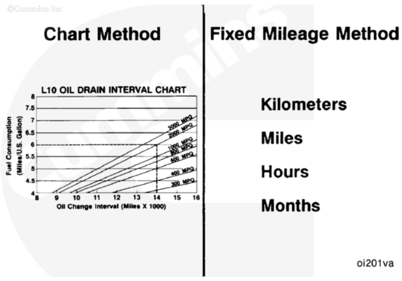
⚠ CAUTION ⚠

The use of a synthetic base oil does not justify extended oil change intervals. Extended oil change intervals can decrease engine life due to factors such as corrosion, deposits, and wear.

⚠ CAUTION ⚠

Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life due to oil degradation processes associated with lightly loaded engines.

Do **not** extend oil and filter change intervals beyond those specified configurations in the maintenance schedule, unless the chart method is used. See the oil interval charts in this section.



⚠ CAUTION ⚠

Inaccurate fuel consumption records can result in incorrect oil change intervals that can result in premature engine wear and/or deposits.

⚠ CAUTION ⚠

Do not extend oil change intervals beyond the maximum point shown when fuel consumption rates are less than the minimum shown. Doing so can result in shortened engine life due to oil degradation processes associated with lightly loaded engines.

Oil change intervals can be determined by one of the following recommended methods:

- Fixed-Hour Method (based on fixed hours or months, whichever occurs first)
- Chart Method (based on known fuel consumption rates, coupled with oil analysis program). The chart method is recommended to provide the lowest total cost of operation, while still protecting the engine.
- Centinel™ System, coupled with an oil analysis program.
- Lubricating Oil Drain Interval Extension Testing. Use the following section of Service Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html), Extended Service Interval Program for Mining Applications and Service Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html), Extended Service Interval Program for

Generator Set Applications. Refer to Section 6. For locomotive applications, use the following section of Service Bulletin 5659817 (/qs3/pubsys2/xml/en/bulletin/5659817.html), Extended Service Interval Program for Locomotive Applications. Refer to Section 6.

The information listed below is required when using the chart method to determine the correct oil and filter change interval for the engine.

- Oil system capacity (sump plus any remote tank volume where oil is continuously circulated with the oil in the engine sump)
- Average fuel consumption rate
- Classification of oil as standard or premium
- Oil analysis to demonstrate the chart method interval is acceptable. See Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html), for guidance on setting up an oil analysis program. Oil samples **must** be taken every 250 hours.

Sump capacity can be determined by knowing the volume of the oil required to touch the high-level mark on the dipstick and the volume of any remote oil tanks on the machine in which oil is continuously circulated. Oil sump capacities are listed in the operation and maintenance manuals for all Cummins® engines. It is imperative that the total system volume be known when using the chart method. Use the following procedure for oil sump capacity information. Refer to Procedure 018-017 in Section V. (/qs3/pubsys2/xml/en/procedures/28/28-018-017-tr.html)

To use the chart method effectively, accurate fuel consumption records **must** be kept and maintained. Fuel consumption information **must** be in liters per hour or gallons per hour units to use the charts. Fuel consumption rates can change over time because of an increasing or decreasing engine duty cycle. Accurate records are essential in determining average fuel consumption during a given oil change interval.

The charts are categorized and labeled by the system capacity for a specific engine family. Select the chart representing the correct oil system capacity for the engine model in question. The left vertical axis of the chart represents fuel consumption in gallons per hour. Determine the intersection point of the fuel consumption rate on the applicable interval curve by drawing a horizontal line. From this point, draw a vertical line down until it intersects with the horizontal axis representing hours. This point represents the acceptable oil change interval.

Centinel™ - With a continuously operating Centinel™ System, the oil drain interval can be extended until the oil analysis requires the oil to be changed or the oil is known to be contaminated. Oil analysis is required at 250 hour intervals with the Centinel™ System and **must** include soot measurement. Fleetguard® Oil Analysis Kit, Part Number CC 2543, meets this requirement. Oil analysis parameters **must** remain within the limits specified in Table 2 of the Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html). Reference the following bulletins for more information on oil sampling and analysis.

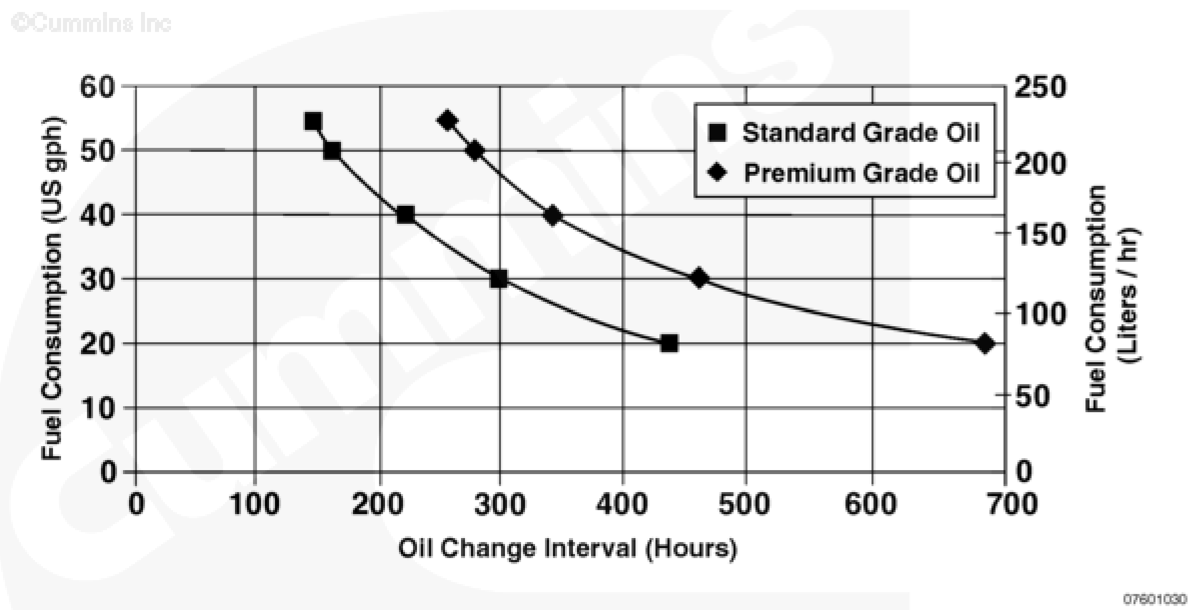
- Fluids for Cummins Products Service Manual, Bulletin 5411406 (/qs3/pubsys2/xml/en/bulletin/5411406.html), Section 4 Engine Oil
- Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html).
- Statistical Analysis of Oil Sample Lead Readings on High Horsepower Engines, Bulletin 2883452 (/qs3/pubsys2/xml/en/bulletin/2883452.html).

Premium oil filters, such as Fleetguard® LF 9000 series filters, or equivalent, are also required when using Centinel™ with spin-on filters.

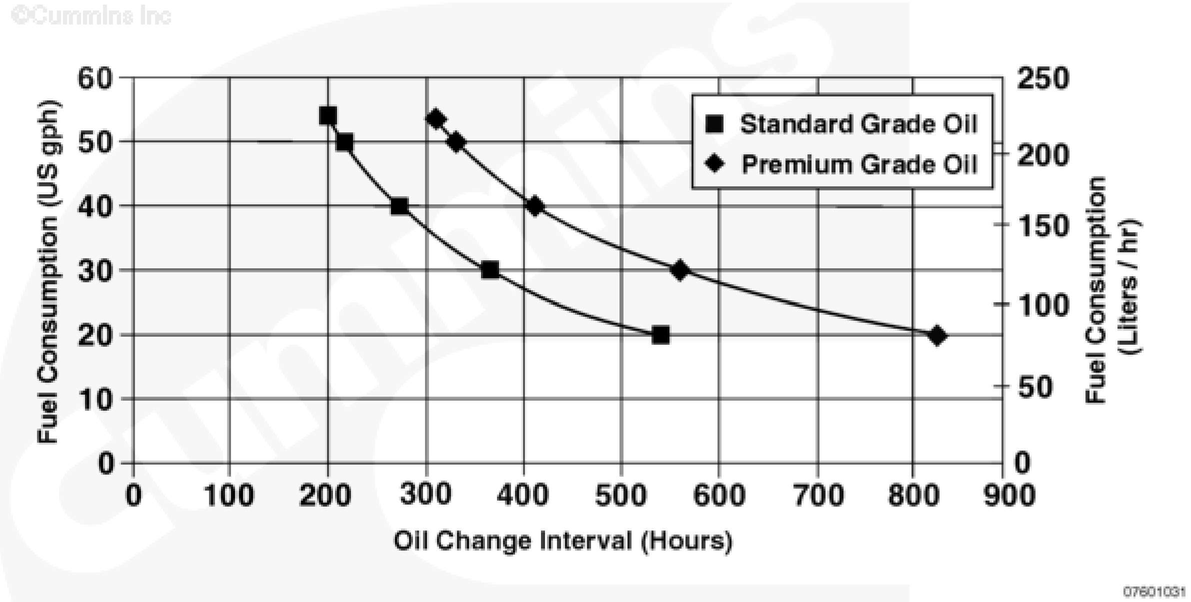
Lubricating Oil Filters

Premium filters are recommended, in conjunction with premium grade oils, when using the premium grade oil curves to determine oil change intervals. Premium filters contain synthetic media materials to provide more efficient filtration for the entire service life and to extend the media life over conventional cellulose media. Premium full-flow filters are made with synthetic media and have the StrataPore™ designation on the outside of the filter. StrataPore™ filters have the efficiency, capacity, and strength needed for this extended service.

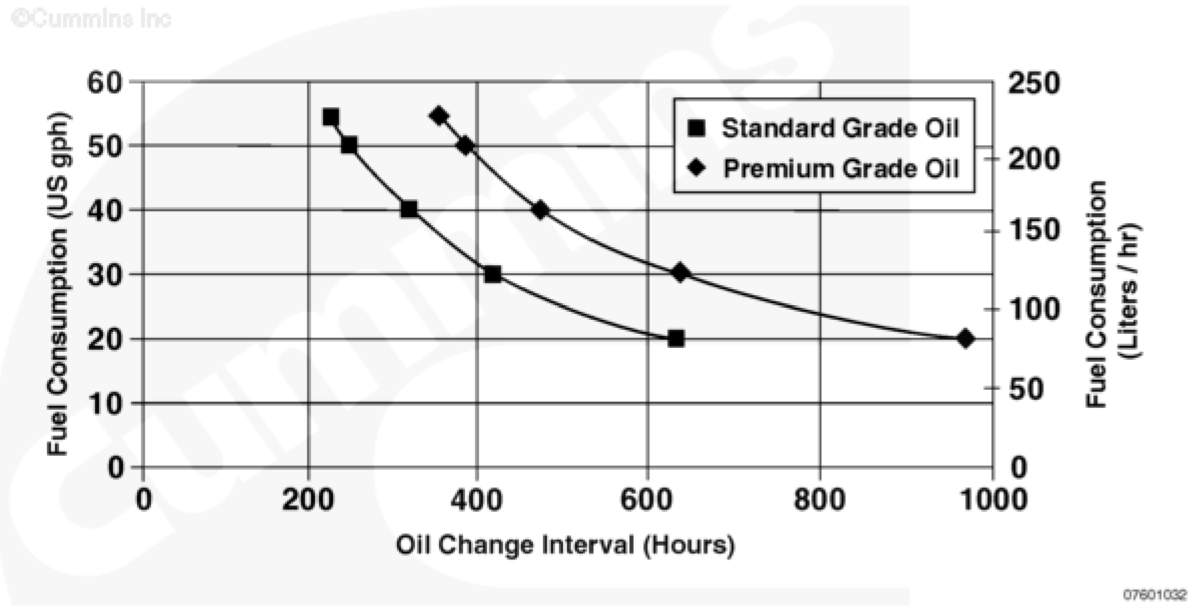
All Applications



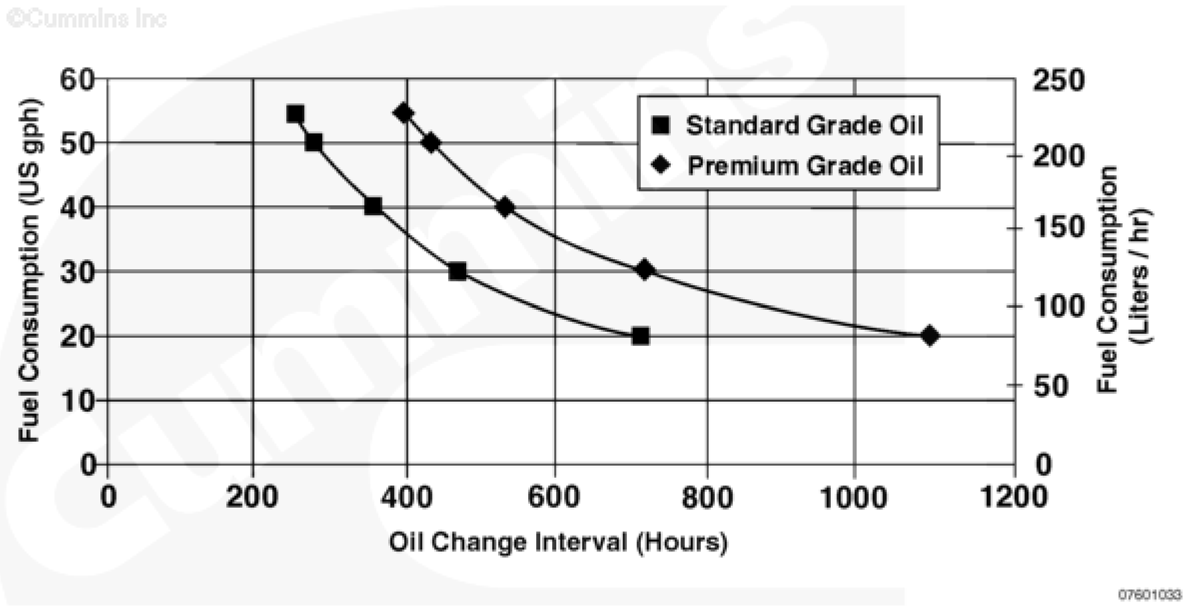
K38 and QSK38 Oil Change Interval - 113 liter [30 gal] Sump (with spin-on or Eliminator™ filters)



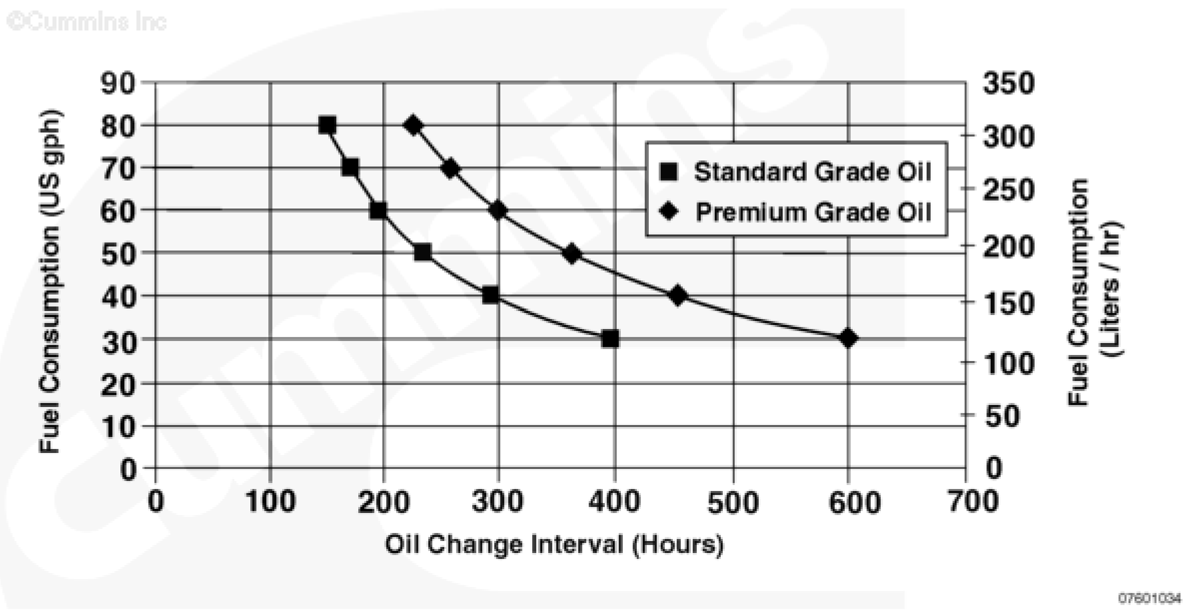
K38 and QSK38 Oil Change Interval - 140 liter [37 gal] Sump (with spin-on or Eliminator™ filters)



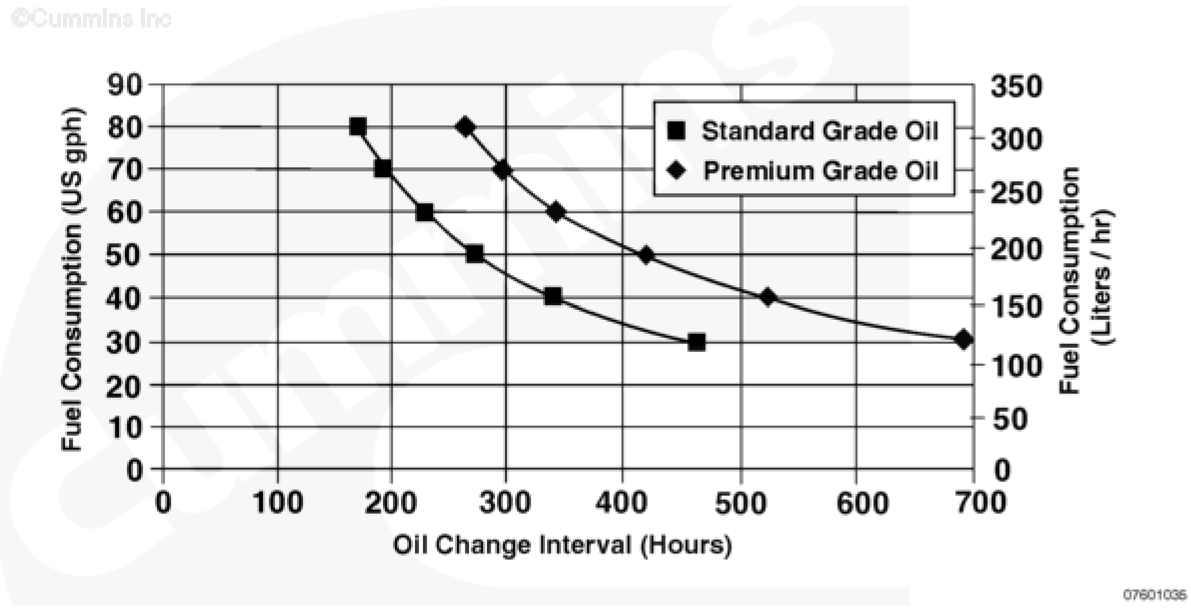
K38 and QSK38 Oil Change Interval - 166 liter [44 gal] Sump (with spin-on or Eliminator™ filters)



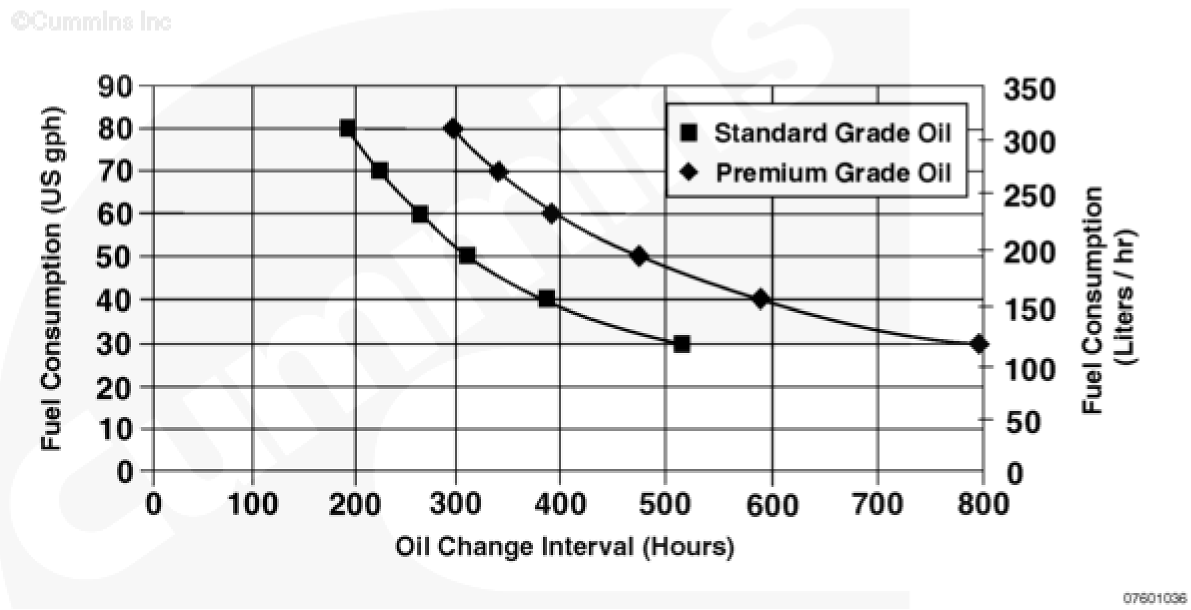
K38 and QSK38 Oil Change Interval - 185 liter [49 gal] Sump (with spin-on or Eliminator™ filters)



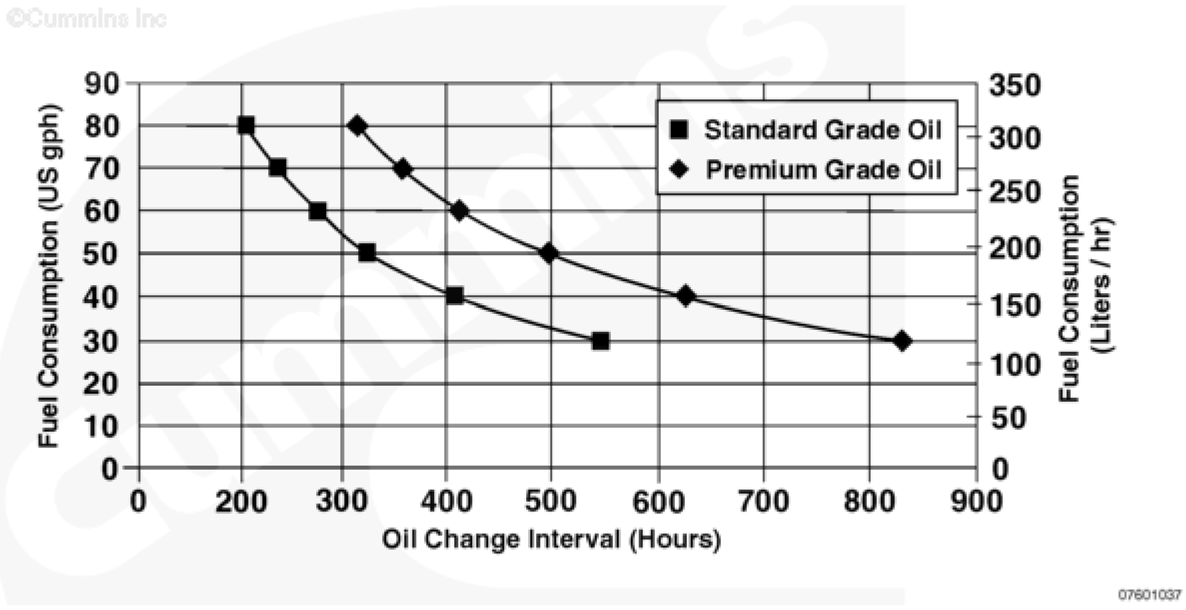
K50 and QSK50 Oil Change Interval - 151 liter [40 gal] Sump (with spin-on or Eliminator™ filters)



K50 and QSK50 Oil Change Interval - 178 liter [47 gal] Sump (with spin-on or Eliminator™ filters)



K50 and QSK50 Oil Change Interval - 204 liter [54 gal] Sump (with spin-on or Eliminator™ filters)



K50 and QSK50 Oil Change Interval - 216 liter [57 gal] Sump (with spin-on or Eliminator™ filters)

Coolant Drain Interval

If following the requirements in the extended service interval program service bulletins, coolant **only** needs to be replaced if indicated through fluid sampling. See Service Bulletin, Extended Service Interval Program for Mining Applications, Bulletin 4388840 (/qs3/pubsys2/xml/en/bulletin/4388840.html) and Service Bulletin, Extended Service Interval Program for Generator Set Applications, Bulletin 5411272 (/qs3/pubsys2/xml/en/bulletin/5411272.html).

Maintenance Intervals for Cooling System up to 568 liters [150 gal]					
	System Size in liters [gal]				
	79-144 [21-30]	117-189 [31-50]	193-284 [51-75]	288-378 [76-100]	382-568 [101-150]
Hours	Units of SCA				
751-1000	25	50	80	75	150
501-750	20	35	60	100	110
251-500	15	25	40	50	75
0-250	10	15	20	25	40

Maintenance Intervals for Cooling System up to 1514 liters [400 gal]					
	System Size in liters [gal]				
	572-757 [151-200]	761-946 [201-250]	950-1135 [251-300]	1139-1325 [301-350]	1329-1514 [351-400]
Hours	Units of SCA				
751-1000	200	250	300	350	400
751-1000	150	190	225	260	300
251-500	100	125	150	175	200
0-250	50	65	75	90	100

Notes:

A. Consult the vehicle equipment manufacturer's maintenance information for total cooling system capacity.
B. When draining and replacing the coolant, **always** pre-charge the cooling system to a SCA level of 1.5 units per gallon. This concentration level **must not** be allowed to go below 1.2 units and **must** be controlled when the level is greater than 3 units. Action needed when the level goes below 1.2 is a filter and liquid pre-charge; from 1.2 to 3.0 units, filter **only**; above 3.0, test at every oil change until the level falls to 3.0 or below.
C. For coolant filter part numbers and SCA capacity. Refer to Procedure 018-024 in Section V. (/qs3/pubsys2/xml/en/procedures/28/28-018-024-om.html)
D. Change coolant filters at each oil change to protect the cooling system. Consult the coolant capacity chart to determine the correct coolant filter for a given cooling system capacity and oil drain interval.